

Cell C4: complete proofs for five and six lines

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Official statement (Cell 4). “Two concrete instances of the next case, $N = d + 2$: five lines in three dimensions and six lines in four. In each, the conjectured optimum repeats two of the coordinate axes. Prove that any 5 lines in $\mathbb{R}^3$ satisfy $S \leq 4\pi$, and that any 6 lines in $\mathbb{R}^4$ satisfy $S \leq 13\pi/2$.”

For unit representatives $x_1, \dots, x_N \in \mathbb{R}^d$, put $g_{ij} = \langle x_i, x_j \rangle$, $A = \sum_{i < j} \arcsin |g_{ij}|$, and $S = \sum_{i < j} \arccos |g_{ij}|$. Since $\theta_{ij} = \pi/2 - \arcsin |g_{ij}|$, the two requested inequalities are both equivalent to $A \geq \pi$.

Elementary lemmas

Proposition 1 (Planar triples and quadruples). *For three lines in a plane, the sum of their pairwise acute angles is at most π . For four lines in a plane, it is at most 2π . Equivalently, their respective A -values are at least $\pi/2$ and π .*

Proof. Represent the three lines by three points on a circle of circumference π ; acute angle is shortest circular distance. If the three successive gaps are all at most $\pi/2$, the three pair distances sum to the circumference π . If one gap is greater than $\pi/2$, the sum of pair distances is twice the sum of the other two gaps, which is less than π . This proves the triple statement. For four lines, sum the triple statement over their four triples. Each pair appears twice, so twice the four-line angle sum is at most 4π . For later use, averaging the three-line $A \geq \pi/2$ bound over the ten triples among five planar lines shows that their A -value is at least $5\pi/3$: every pair occurs in three triples. $\square$

Proposition 2 (Orthogonal basis). *If $e_1, \dots, e_d$ are an orthonormal basis and z is a unit vector, then $\sum_{j=1}^d \arcsin |\langle z, e_j \rangle| \geq \pi/2$.*

Proof. For $a, b \geq 0$ with $a^2 + b^2 \leq 1$ one has

$$\arcsin a + \arcsin b \geq \arcsin \sqrt{a^2 + b^2}.$$

If the left side is already at least $\pi/2$, this is immediate. Otherwise, taking sines reduces the claim to $\sqrt{(1-a^2)(1-b^2)} \geq ab$, which follows from $a^2 + b^2 \leq 1$. Combine the d coordinates successively and use $\sum_j |\langle z, e_j \rangle|^2 = 1$. $\square$

Proposition 3 (Orthogonality saturation). *Fix N lines in $\mathbb{R}^d$ and let S be their total acute-angle sum. There exists a maximizer of S such that, for every line ℓ_i , either it is orthogonal to all other lines or its orthogonality neighbors span a space of dimension at least $d - 1$.*

Proof. The product of projective spheres is compact, so a maximizer exists. Among all maximizers choose one with the largest possible number of orthogonal pairs. Let W be the span of the representatives of the lines orthogonal to ℓ_i . Suppose $\dim W \leq d-2$ and that ℓ_i is not orthogonal to all the others. Then $W^\perp$ has dimension at least two, and there is a great circle in $W^\perp$ through a unit representative x_i on which the inner product with some non-neighbor is nonconstant. Vary only x_i on this circle.

On an interval with no new orthogonality event, each term $\arccos |\langle x_i(t), x_j \rangle|$ is convex. To see this, write its inner product, up to sign, as $r \cos(t - \phi)$, with $0 \leq r \leq 1$ and $\cos(t - \phi) > 0$. For $r < 1$ its second derivative is

$$\frac{r(1-r^2)\cos(t-\phi)}{(1-r^2\cos^2(t-\phi))^{3/2}} \geq 0.$$

For $r = 1$ the function is piecewise linear with a convex corner at coincidence. Terms with $r = 0$ are constant. Thus the sum is convex between consecutive orthogonality events. One endpoint of the event interval containing x_i has value at least its initial value. Since the initial configuration was a global maximizer, the endpoint is another maximizer. All old orthogonality pairs involving i persist, since the circle lies in $W^\perp$, and at least one new pair appears. This contradicts the choice of maximizer. $\square$

The same argument has a strict form that we will use only to describe equality. At *any* maximizer, suppose an orthogonality-neighbor span W has dimension at most $d-2$. On every two-dimensional circle $U \subset W^\perp$ through x_i , a non-neighbor y with $0 < \|\text{proj}_U y\| < 1$ would make its angle term strictly convex between events, contradicting maximality. Thus every non-neighbor belongs to every such U . If $\dim W^\perp = 2$, all non-neighbors lie in $W^\perp$; if $\dim W^\perp \geq 3$, they lie on ℓ_i , the intersection of these circle planes. Hence the configuration splits into orthogonal subfamilies, unless $W = 0$ and all lines coincide. The latter is not a maximizer for $N \geq 2$ and $d \geq 2$.

Five lines in three dimensions

Proposition 4. *For any five lines in $\mathbb{R}^3$, $A \geq \pi$, or equivalently $S \leq 4\pi$.*

Proof. It suffices to prove the bound for a maximizing configuration chosen as in the orthogonality-saturation proposition. Let H be its graph of orthogonal pairs. If a vertex is adjacent to all four others, those four lines lie in a plane; their angle sum is at most 2π , and the four angles involving the universal vertex total 2π .

Otherwise the neighbors of every vertex span a plane, so H has minimum degree at least two. If H contains a triangle, its three lines form an orthonormal basis. The orthogonal-basis proposition applied to each of the remaining two lines gives $A \geq \pi$.

If H is triangle-free, minimum degree two on five vertices forces $H = C_5$ or $H = K_{2,3}$. Indeed, degree two everywhere gives C_5 ; if some vertex has degree three, its three neighbors are mutually nonadjacent, and the fifth vertex must meet all three to give them degree at least two, giving $K_{2,3}$. In the latter case the two parts span orthogonal subspaces. If the part of size two consists of independent lines, the three lines in the other part coincide and contribute $3\pi/2$ to A . Otherwise the two lines coincide, contributing $\pi/2$, while the other three are planar and contribute at least another $\pi/2$.

It remains to treat C_5 . Label its edges 12, 23, 34, 45, 51. Choose coordinates with $x_1 = e_1$ and $x_2 = e_2$. Since $x_2 \perp x_3$ and $x_1 \perp x_5$, write, up to signs irrelevant to the absolute inner products,

$$x_3 = (\cos t, 0, \sin t), \quad x_5 = (0, \cos s, \sin s), \quad 0 \leq t, s \leq \pi/2.$$

The line x_4 is perpendicular to both x_3, x_5 . Put $D = \sqrt{1 - \sin^2 t \sin^2 s}$. If $D = 0$, then $x_3 = x_5 = e_3$ as lines and x_1, x_2, x_3 are an orthonormal basis, a case already handled. Otherwise, the five nonzero-potential pairs are the nonedges of C_5 , and direct cross-product calculation gives

$$A = \pi - t - s + u(t, s) + v(t, s) + \gamma(t, s) = \pi + F(t, s), \quad F(t, s) := u(t, s) + v(t, s) + \gamma(t, s) - t - s,$$

where

$$u = \arctan(\tan t \cos s), \quad v = \arctan(\tan s \cos t), \quad \gamma = \arcsin(\sin t \sin s).$$

The expression is interpreted continuously at endpoints. We show $F \geq 0$. For fixed $s \in [0, \pi/2)$, differentiation gives

$$\frac{\partial F}{\partial t} = \frac{\cos s}{1 + \sin t \sin s} + \frac{\cos t \sin s}{\sqrt{1 - \sin^2 t \sin^2 s}} - 1.$$

Both nonconstant terms decrease with t . For the second,

$$\frac{d}{dt} \left(\frac{\cos t}{D} \right) = -\frac{\sin t \cos^2 s}{D^3} \leq 0.$$

Hence F is concave in t . But $F(0, s) = F(\pi/2, s) = 0$, so $F(t, s) \geq 0$. The case $s = \pi/2$ follows by continuity. Therefore $A \geq \pi$. $\square$

Six lines in four dimensions

Proposition 5. *Any four lines in $\mathbb{R}^3$ satisfy $A \geq \pi/2$.*

Proof. Choose a maximizing configuration for their acute-angle sum with the largest number of orthogonal pairs. If one line is orthogonal to the other three, the latter are planar, so their A -value is at least $\pi/2$. Otherwise saturation gives minimum degree two in the orthogonality graph on four vertices. If it contains a triangle, that triangle is an orthonormal basis and the fourth line contributes at least $\pi/2$ by the orthogonal-basis proposition. If it is triangle-free, it must be $K_{2,2}$: the two parts span orthogonal spaces, and since their dimensions sum to at most three, at least one part is a pair of coincident lines, contributing $\pi/2$ to A . $\square$

Proposition 6. *For any six lines in $\mathbb{R}^4$, $A \geq \pi$, or equivalently $S \leq 13\pi/2$.*

Proof. The same saturation proposition gives a useful restriction. At a maximizing configuration of six lines in $\mathbb{R}^4$, a universal orthogonality vertex reduces the bound to the five-line theorem above: the other five lines lie in its orthogonal $\mathbb{R}^3$, while its five incident angles total $5\pi/2$. Four mutually orthogonal lines also settle the case, by applying the orthogonal-basis proposition to each of the other two lines.

If neither occurs, let J be the graph of nonorthogonal pairs. Every orthogonality neighborhood spans a three-dimensional space, so every vertex of J has degree at most two. If J is connected, it is a path or a cycle on six vertices. A path is impossible: the Gram matrix, in path order, is an irreducible symmetric tridiagonal matrix, whose eigenvalues are simple (the first coordinate of an eigenvector determines all the others by the three-term recurrence). Its nullity is therefore at most one, whereas six vectors in $\mathbb{R}^4$ have Gram nullity at least two. If J is disconnected, its components are mutually orthogonal families. For any component of at most five lines with span rank r , the preceding planar bounds, the four-line proposition, and the five-line theorem show directly that its A -value is at least $(n-r)\pi/2$: check $r = 1$ by counting coincident pairs, $r = 2$ by the planar bounds and the five-line triple average, and $r = 3$ by the four- and five-line theorems. A component of

rank 4 leaves no room for any other nonempty orthogonal component. Summing over components gives $A \geq (6 - 4)\pi/2 = \pi$. It remains to handle $J = C_6$. This case also follows from the general cycle lemma in `p1_c5.tex`. The two alternating triples of its vertices are each pairwise orthogonal. Denote their unit vectors by u_1, u_2, u_3 and v_1, v_2, v_3 , relabeling so that $u_i \perp v_i$ for $i = 1, 2, 3$ and the six other cross pairs are the edges of J . Complete u_1, u_2, u_3 to an orthonormal basis of $\mathbb{R}^4$, so $u_i = e_i$. Orienting coordinate axes and lines does not change A . We may therefore write

$$v_1 = (0, C, Pc, Ps), \quad v_2 = (D, 0, Ws, -Wc),$$

where $P = \sin p$, $C = \cos p$, $W = \sin w$, $D = \cos w$, $c = \cos q$, $s = \sin q$, and $p, w, q \in [0, \pi/2]$. Indeed, the first expression parameterizes a unit vector orthogonal to e_1 ; the second parameterizes a unit vector orthogonal to both e_2 and v_1 . For the strict C_6 pattern all six cross inner products are nonzero, so $C, D, c, s > 0$. Boundary cases follow by continuity or have already been covered by a graph with more orthogonality edges.

The vector v_3 is orthogonal to e_3, v_1, v_2 . Solving these three linear equations and normalizing shows that the absolute values of its first two coordinates are

$$X = \frac{WCc}{H}, \quad Y = \frac{PDs}{H}, \quad H^2 = C^2D^2 + P^2D^2s^2 + W^2C^2c^2.$$

The six nonzero-potential pair terms are consequently

$$A = \arcsin C + \arcsin D + \arcsin(Pc) + \arcsin(Ws) + \arcsin X + \arcsin Y = \pi + F(q),$$

where

$$F(q) = \arcsin(Pc) + \arcsin(Ws) + \arcsin X + \arcsin Y - p - w.$$

We prove $F(q) \geq 0$. Its endpoint values are $F(0) = F(\pi/2) = 0$. Set

$$a = \sqrt{1 - P^2c^2}, \quad b = \sqrt{1 - W^2s^2}, \quad M = \frac{CD}{H^2}.$$

Direct differentiation, using $1 - X^2 = D^2a^2/H^2$ and $1 - Y^2 = C^2b^2/H^2$, gives

$$F'(q) = \frac{c(W + PM)}{b} - \frac{s(P + WM)}{a}.$$

For $0 < p, w < \pi/2$, the sign of $F'(q)$ is the sign of $R(q) - 1$, where

$$R(q) = \frac{c}{s} \frac{a}{b} \frac{W + PM}{P + WM}.$$

The first two factors have a nonincreasing product: multiplying its logarithmic derivative by sc yields

$$-1 + \frac{P^2s^2c^2}{a^2} + \frac{W^2s^2c^2}{b^2} \leq 0,$$

because $a^2 = C^2 + P^2s^2 \geq P^2s^2$ and $b^2 = D^2 + W^2c^2 \geq W^2c^2$. The last factor is also nonincreasing. Indeed $H^2 = D^2 + (W^2 - P^2)c^2$, so

$$M' = \frac{2CD(W^2 - P^2)sc}{H^4},$$

and the logarithmic derivative of $(W + PM)/(P + WM)$ is

$$\frac{M'(P^2 - W^2)}{(W + PM)(P + WM)} = -\frac{2CD(W^2 - P^2)^2sc}{H^4(W + PM)(P + WM)} \leq 0.$$

Thus R decreases from $+\infty$ at $q = 0$ to 0 at $q = \pi/2$. The derivative F' changes sign only from positive to negative, so F has its minimum at an endpoint. Both endpoint values are zero; hence $A \geq \pi$. Cases with p or w on the boundary follow by continuity. This completes the six-line proof. $\square$

The six-line case also follows from the cycle lemma of Cell 5 (applied with $m = 6$), which gives a second, independent proof.

Sharpness

Three planar lines with directions $0, \pi/3, 2\pi/3$ have three pairwise acute angles $\pi/3$, hence contribute $\pi/2$ to A . Four planar lines with directions $0, \pi/2, \delta, \delta + \pi/2$ have $A = \pi$ for every $0 \leq \delta \leq \pi/2$. For five lines in $\mathbb{R}^3$, put the planar triple in one two-dimensional summand and repeat the perpendicular line twice; their contributions are $\pi/2 + \pi/2 = \pi$. Alternatively, use the planar quadruple and one perpendicular line. For six lines in $\mathbb{R}^4$, put one such planar triple in each of two mutually orthogonal planes. Their contributions again sum to π . Thus both upper bounds are attained.

Equality configurations

Here is an exact description, with repeated lines allowed. Let D be a pair of coincident lines; let T be any three lines in a plane whose three successive gaps on the projective circle are all at most $\pi/2$; let Q be any four lines in a plane that are the union of two orthogonal pairs. The symbol $+$ below denotes an orthogonal direct sum of the spans of the indicated blocks, and 1 denotes a single line. Up to an orthogonal transformation and relabeling, all equality configurations are

(N, d)	blocks
$(5, 3)$	$Q + 1$ or $T + D$
$(6, 4)$	$Q + 1 + 1$ or $T + D + 1$ or $T + T$.

Each displayed family attains $A = \pi$, by the planar bounds with equality and by $A(D) = \pi/2$.

For completeness, we justify that the list is exhaustive. Equality in the orthogonal-basis proposition holds precisely when the extra vector has at most two nonzero basis coordinates: combining two positive coordinates is strict whenever their squared sum is less than one. Equality for a planar triple is exactly the gap condition in the definition of T . For four cyclically ordered planar lines, write the four consecutive gaps as a, b, c, d , with $a + b + c + d = \pi$. Equality in the four-line bound forces equality in each of its four triple bounds, hence

$$a + b, b + c, c + d, d + a \leq \pi/2.$$

The four left sides sum to 2π , so all are $\pi/2$; opposite lines form two orthogonal pairs. This is exactly Q .

An equality configuration is a global maximizer. If it is not an orthogonal sum, the strict form of orthogonality saturation gives the same minimum-degree conclusions for its own orthogonality graph as those used above. For four lines in $\mathbb{R}^3$, its graph is then a triangle or $K_{2,2}$; the latter is an orthogonal sum, and equality in the former puts the fourth line in a coordinate plane, leaving a basis line orthogonal to the other three. Thus every four-line equality case is $T + 1$.

For five lines in $\mathbb{R}^3$, the $K_{2,3}$ case is an orthogonal sum and the exact C_5 case is strict: with $0 < t, s < \pi/2$, the function $F(t, s)$ above is strictly concave in t and is positive between its zero

endpoints. In the orthogonal-basis case, the two extra lines each have support of size at most two and are orthogonal to each other. If their supports are disjoint or equal, they split off an orthogonal coordinate block; supports meeting in exactly one coordinate have nonzero inner product. Hence this case also splits orthogonally. Decomposing repeatedly, and using the sharp component bounds in ranks one, two, and three, leaves exactly $Q + 1$ or $T + D$.

For six lines in $\mathbb{R}^4$, the exact C_6 case is strict because $R(q)$ is strictly decreasing in the interior, so $F(q) > 0$ there. The path is impossible and a disconnected nonorthogonality graph already gives an orthogonal sum. If four lines form an orthonormal basis, equality requires the other two to be orthogonal and each to have support in at most two basis coordinates. Their supports are either disjoint or equal; supports meeting in only one coordinate would have nonzero inner product. Thus this case also splits. Finally, the universal-vertex case splits by definition. Every equality case is therefore an orthogonal sum of components of sizes at most five. The component inequalities used above imply that equality requires total span rank four and equality in each component. The only positive-defect block types needed are D (defect one), T (defect one), and Q (defect two); the four-line and five-line equality classifications just proved split all other blocks further. A zero-defect component at equality has $A = 0$, so its lines split into orthogonal singletons. Counting six lines and rank four gives exactly the three displayed families.

An independent short estimate

Proposition 7. *For $(N, d) = (5, 3)$ or $(6, 4)$, the requested inequality holds if $|g_{ij}| \leq 1/2$ for every $i \neq j$. In the first case the stronger bound $A \geq 10\pi/9$ holds.*

Proof. The function $f(t) = \arcsin(t)/t^2$ decreases on $(0, 1/2]$. Indeed, the sign of $f'(t)$ is the sign of $t/\sqrt{1-t^2} - 2\arcsin t$, which is negative there because $t/\sqrt{1-t^2} \leq 2t/\sqrt{3} < 2t \leq 2\arcsin t$. Consequently,

$$\arcsin t \geq \frac{2\pi}{3}t^2 \quad (0 \leq t \leq 1/2).$$

Let $T = \sum_i x_i x_i^\top$. It is positive semidefinite, $\text{tr } T = N$, and Cauchy–Schwarz on its at most d nonzero eigenvalues gives

$$N + 2 \sum_{i < j} g_{ij}^2 = \text{tr}(T^2) \geq \frac{N^2}{d}.$$

Thus

$$A \geq \frac{2\pi}{3} \sum_{i < j} g_{ij}^2 \geq \frac{\pi N(N-d)}{3d}.$$

This last expression is $10\pi/9$ for $(N, d) = (5, 3)$ and π for $(N, d) = (6, 4)$. □

The hypothesis is exactly that every acute line angle is at least $\pi/3$. This gives a separate check on that part of the configuration space.